

Peak DRAM temperature (°C)

100

90

80

70

60

0%

1

2

3

4

5

6

7

66%

8

9

10

11

12

100%

Number of FeRAM dies replacing DRAM (of 12-high stack)

- Peak DRAM temp (measured)
- Peak FeRAM temp (no refresh; retention 4-8 yr)
- Decode throughput (timing-aware BW model)

DRAM thermally constrained (> 85 °C)

XBM operating point (N = 4)  
DRAM: 85.2 °C — at the 85 °C limit  
throughput 87 %

90 °C DRAM limit

85 °C DRAM limit

all-FeRAM: no DRAM for the mutable KV-cache → throughput collapses

Decode throughput (% of all-DRAM)

100

80

60

40

20

0

Timing-aware BW model: FeRAM read BW =  $0.60 \times \text{DRAM (tRC + destructive-read write-back)}$ ; mutable KV-cache needs  $\geq 2$  DRAM dies → knee at N = 10.

Temperatures measured from PACT; throughput modelled.